An Explicit GCD Evaluation for Three-Direction Torus Arrangements

Rushikesh Murlidhar Walke
Independent Researcher, Maharashtra, India
ORCID: https://orcid.org/0009-0006-6749-5660

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Abstract

We explicitly evaluate established toric-arrangement counting principles for a family of closed curves on the two-dimensional torus. The vertical coordinates form the union of two common-origin cyclic grids of orders N and R; the other two families are the level sets of $x + y$ and $x - y$ on grids of orders Y and K. A finite-group calculation gives the number of triple intersections, and a cellularity argument justifies the use of Euler's relation. For $R = 12$, $Y = 27$, and $K = 60$, the face count is $87N + 4248 - 84\gcd(N, 12) - 3\gcd(N, 1080)$. In particular, it is 6264 at $N = 27$. The result concerns geometric faces and does not require a sampled trajectory or numerical experiment. We give an elementary proof and identify the same expression as an arithmetic-Tutte specialization.

1 Relation to existing work

Chamber counting for toric arrangements is established theory. Ehrenborg, Readdy and Slone give a general count from the intersection poset under an open-ball hypothesis [2, Theorem 3.6]. For two-dimensional arrangements, the identity $F = \sum_{j\geq 2}(j-1)t_j$, with t_j counting intersections of j curves, appears in the preprint version of Chandrasekhar and Deshpande [1, Lemma 2.4]. Our $F = P - T$ is its double/triple-point case. Moci's arithmetic multiplicities and compact-torus chamber formula [3, Section 2.2, Lemma 5.4 and Corollary 5.16] also yield the expression evaluated here.

The purpose of this note is to display the gcd reduction for the specified common-origin grids, including duplicate vertical levels, and provide a self-contained proof and parameterized example. Neither Euler's relation nor a new general chamber-count principle is claimed as a contribution. The precise literature connections are made explicit below; finding no identical printed numerical example would not establish priority.

The author's earlier expository note [4, pp. 2–5, Theorem 1 and Corollary 1] counts pairs on finite cyclic grids subject to sum and difference constraints, using Smith normal form. Its formula specializes exactly to the triple-point count used here, as shown in Section 4. The present note adds the torus arrangement, the cellularity argument, and the passage from point incidences to geometric faces. The two notes therefore share a finite-grid counting component while counting different final objects. Neither note claims a new general counting theory.

2 Setup and scope

For a positive integer m, let

$$G_m = \{j/m + \mathbb{Z} : 0 \le j < m\} \subset \mathbb{R}/\mathbb{Z}.$$

All grids have origin zero; independently rotated grids are outside the stated assumptions. Fix positive integers N, R, Y, K. On $\mathbb{R}^2/\mathbb{Z}^2$ consider the curves

$$x = u \quad (u \in U := G_N \cup G_R), \qquad x + y = a \quad (a \in G_Y), \qquad x - y = b \quad (b \in G_K),$$

where equations are understood modulo one. A vertical curve occurring in both grids is included only once. Each individual curve is an embedded circle. Curves in the same family are disjoint, and curves from different families meet transversely.

Vertices are distinct intersection points. Edges are the arcs between successive vertices on each curve, allowing a loop edge when there is only one vertex on a curve. Faces are connected components of the complement. Write their counts as V, E, F, and write T for the number of points on three curves. There are no points on more than three distinct curves.

The problem is one of combinatorial topology. Euler's relation below is the surface formula for a graph embedding, not the Euler equations of fluid mechanics. We make no claim here about orbital sampling, error detection, astronomical accuracy, or the number of distinct labels after a relabelling of the faces.

3 Face-count theorem

Put

$$g = \gcd(N, R), \quad v = N + R - g, \quad d = \gcd(Y, K), \quad L = \operatorname{lcm}(Y, K), \quad P = v(Y + K) + 2YK.$$

Theorem 1. *For the arrangement just defined, every face is an open disk. Moreover,*

$$T = d(\gcd(N, 2L) + \gcd(R, 2L) - \gcd(g, 2L)), \tag{1}$$

$$V = P - 2T, \qquad E = 2P - 3T, \qquad F = P - T. \tag{2}$$

These formulas hold for all positive integers N, R, Y, K.

Lemma 1. *If the vertical family is a single grid G_n, its triple-point count is*

$$TT(n; Y, K) = d \gcd(n, 2L).$$

Proof. The homomorphism

$$G_Y \times G_K \longrightarrow G_L, \qquad (a, b) \longmapsto a + b$$

is surjective. Indeed, $L/Y = K/d$ and $L/K = Y/d$ are coprime, so Bezout's identity shows that $1/L$ belongs to $G_Y + G_K$. Thus this sum is G_L. Its kernel has size $YK/L = d$, so each element of G_L has exactly d preimages.

A fixed vertical coordinate u supports a triple point exactly when $a + b = 2u$ for some $(a, b) \in G_Y \times G_K$. Each such pair gives the unique point $(u, a - u)$, and distinct pairs give distinct points at that vertical coordinate. Thus there are d triple points at u if $2u \in G_L$, and none otherwise. The condition $2u \in G_L$ is equivalent to $u \in G_{2L}$. Finally, $G_n \cap G_{2L} = G_{\gcd(n,2L)}$, which proves the claim. □

Lemma 2. *The faces of the arrangement are open disks.*

Proof. Lift all the curves to affine lines in $\mathbb{R}^2$. Each chamber of the lifted complement is an intersection of open strips, hence is convex. In particular it lies between two consecutive vertical lines and between two consecutive lines of the form $x + y =$ constant. These two strip directions are independent, so the chamber is bounded. The line arrangement is locally finite, so each chamber is a bounded convex open polygon and is homeomorphic to an open disk.

If a chamber intersects one of its integer translates, the two are the same connected component of the lifted complement. A nonzero translation cannot preserve a bounded nonempty chamber. Consequently the covering projection identifies no two points of its interior. Its image is a whole component of the torus complement: a path in that component lifts to a path remaining in the chamber. Thus each face is homeomorphic to a lifted chamber. Boundary identifications can occur and do not invalidate cellularity of this graph embedding. □

Proof of Theorem 1. Since $G_N \cap G_R = G_g$, there are v distinct vertical curves. A vertical curve meets each sum curve once and each difference curve once. A pair $x+y = a$, $x-y = b$ has exactly two solutions: the equation $2x = a + b$ has two roots modulo one, and each root determines y. Therefore the total pairwise intersection count, with multiplicity, is P.

Apply Lemma 1 to G_N, G_R, and their intersection G_g. Inclusion–exclusion gives equation (1). If D denotes the number of double points, then $P = D + 3T$ and $V = D + T$, giving $V = P - 2T$.

Double and triple vertices have degrees four and six respectively. Every curve has a vertex because the other two families are nonempty. The degree sum, counting a loop twice, gives $2E = 4D + 6T$, hence $E = 2P - 3T$. Lemma 2 makes this a finite cellular embedding on the torus. Euler's relation $V - E + F = 0$ now gives $F = E - V = P - T$. □

4 Connection with the earlier cyclic-grid note

For comparison with [4, Eq. (1)], write

$$C(a, b; Y, K) = \#\{(u, h) \in G_a \times G_b : u + h \in G_Y,\ u - h \in G_K\}.$$

The earlier note's union count is

$$Q(a, b; Y, K; r) = C(a, b; Y, K) + C(r, b; Y, K) - C(\gcd(a, r), b; Y, K).$$

Here $d = \gcd(Y, K)$ and $L = \operatorname{lcm}(Y, K)$ retain the notation of this note; the earlier note uses different letters for some of these quantities.

At a triple point, $2y = (x + y) - (x - y) \in G_L$, so $y \in G_{2L}$. Consequently, restricting the second coordinate to G_{2L} loses no triple points, and there are exact identifications

$$TT(n; Y, K) = C(n, 2L; Y, K), \qquad T = Q(N, 2L; Y, K; R).$$

Conversely, each pair in the specified C or Q set is a triple point of the corresponding arrangement, so these identifications are bijective.

To display the specialization of the earlier Smith-normal-form formula, put $q = \gcd(n, 2L)$, $n = qa$, $2L = qb$, and $M = \operatorname{lcm}(n, 2L) = qab$. Then $\gcd(a, b) = 1$, and the integer matrix in [4, Eqs. (2)–(4)] is

$$A = \begin{pmatrix} Yb & Ya \\ Kb & -Ka \end{pmatrix}.$$

Its invariant factors are

$$s_1 = d, \qquad s_2 = \frac{2YK\,ab}{d} = 2L\,ab = Mb.$$

Since $d \mid M$, the earlier formula gives

$$C(n, 2L; Y, K) = \frac{n(2L)}{M^2} \gcd(M, s_1) \gcd(M, s_2) = \frac{n(2L)}{M} d = dq.$$

Its inclusion–exclusion formula [4, Corollary 1, Eq. (8)] therefore gives exactly equation (1). This is an algebraic specialization of the earlier exposition, not a separate originality claim for the triple-count arithmetic. The earlier note does not count complementary faces of a torus arrangement; the geometric argument above is needed to pass from T to F.

5 An arithmetic-Tutte specialization

This section is a direct specialization of the cited general results, not a further general counting theorem. Use the indexed list

$$X = ((N, 0), (R, 0), (Y, Y), (K, -K)).$$

Its character kernels give exactly the union of curves studied here. For a rank-two sublist A, its arithmetic multiplicity $m(A)$ is the gcd of its nonzero absolute two-by-two determinants. The region count is the arithmetic-Tutte evaluation

$$M_X(1, 0) = \sum_{\substack{A \subseteq X \\ r(A)=2}} (-1)^{|A|-2} m(A) \qquad [3].$$

Here $r(A)$ is the rank over $\mathbb{Q}$ of the indexed character sublist. Subsets are subsets of indexed positions, even if $N = R$. Write $S(n) = \gcd(nY, nK, 2YK) = d \gcd(n, 2L)$. The following evaluation is our algebraic substitution into that known formula:

$$\begin{aligned} \text{pairs:} \quad & (N + R)(Y + K) + 2YK, \\ \text{triples:} \quad & - g(Y + K) - S(N) - S(R), \\ \text{four positions:} \quad & + S(g). \end{aligned}$$

Their sum is $P - T$. Thus the explicit formula is already a consequence of established arithmetic-arrangement theory. The elementary proof above avoids needing that framework to verify this particular evaluation.

6 The parameters $R = 12$, $Y = 27$, $K = 60$

Here $d = 3$ and $L = 540$, so

$$TT(n; 27, 60) = 3 \gcd(n, 1080).$$

Equivalently, writing $\nu_p(n)$ for the exponent of a prime p in n,

$$TT(n; 27, 60) = 2^{\min(\nu_2(n),3)} 3^{\min(\nu_3(n)+1,4)} 5^{\min(\nu_5(n),1)}.$$

This equivalence follows from $1080 = 2^3 \cdot 3^3 \cdot 5$; it does not depend on extrapolating a finite numerical pattern. The function TT itself is not multiplicative in the standard arithmetic sense, since $TT(1) = 3$. Its normalization $TT(n)/3 = \gcd(n, 1080)$ is multiplicative.

Corollary 1. *For every positive integer N, with the other three parameters fixed as above,*

$$\boxed{F(N) = 87N + 4248 - 84 \gcd(N, 12) - 3 \gcd(N, 1080).}$$

Proof. Set $g = \gcd(N, 12)$. Since g divides both 12 and 1080, the triple count is $T = 3 \gcd(N, 1080) + 36 - 3g$. Also $P = 87(N + 12 - g) + 3240$. Subtracting proves the result. □

At $N = 27$ one obtains

$$v = 36, \quad T = 108, \quad P = 6372, \quad (V, E, F) = (6156, 12420, 6264).$$

For the immediate integer neighbours,

$$F(26) = 6336, \qquad F(27) = 6264, \qquad F(28) = 6336.$$

Thus $N = 27$ is a strict local minimum relative to these neighbours. It is not a local maximum. These statements do not assert any physical optimality or lack of arithmetic structure.

For clarity, $TT(28; 27, 60) = 12$ counts triples on the single grid G_{28}, whereas $T = 36$ counts triples on $G_{28} \cup G_{12}$. These quantities must not be interchanged.

7 What the count does not identify automatically

The number of geometric faces need not equal the number of distinct tuples produced by labelling them. For example, $N = R = Y = K = 1$ gives $T = 1$, $P = 4$, and $(V, E, F) = (2, 5, 3)$, while the four one-bin raw labels are constant. A claim equating faces with a particular labelled encoding requires a separate injectivity or connectedness argument. A further claim that a finite trajectory visits every face requires additional analysis. Neither identification is used in Theorem 1.

8 Supplementary implementation

The companion file `anc/hv27_torus_counts.py` provides the gcd formula and a rational vertex-enumeration method using only Python's standard library. The latter counts incidences at exact rational vertices and applies the cellular Euler relation justified above. Comparing the two implementations is a consistency check; the all-parameter theorem is established by the proof, not by such comparisons.

The accompanying `anc/run_hv27_checks.py` defines a finite validation suite. Author-supplied reports dated 6 November 2026 record a Windows CPython 3.14.7 run with 284 completed checks, all passing: two reference examples, 26 default-parameter cases with $15 \leq N \leq 40$, eight nondefault-parameter examples, 200 pure-grid triple-count comparisons with $1 \leq n \leq 200$, and 48 invalid-input cases. The pure-grid test uses the integer congruence

$$K(2iY - jn) \equiv 0 \pmod{nY}, \qquad 0 \leq i < n, \quad 0 \leq j < Y,$$

instead of the companion's rational vertex construction.

The supplied reports are included as `anc/validation/RESULTS.json` and `anc/validation/RESULTS.txt`. Their recorded checks agree, and the recorded core and runner SHA-256 hashes match the distributed source files. These are inspected author-supplied runtime records, not an independent re-execution by the preparation assistant. They do not record a terminal process exit code. Earlier archived logs concern other source versions and are not used to validate this release.

These finite checks support implementation consistency. In particular, the vertex enumerator obtains F from $E - V$; it does not independently enumerate the complement's faces. The cellularity proof and the all-parameter theorem remain mathematical arguments, rather than conclusions drawn from a finite run.

Assistance and author review

ChatGPT (OpenAI) and Claude (Anthropic) were used for AI assistance in the research and preparation of this note. Across the workflow, this assistance included calculations, proof arguments, mathematical exposition, literature comparison, and supplementary source code. It was not limited to language editing. Historical computation records and the new author-supplied validation reports were read as data. No research code or proof checker was executed by the preparation assistant.

The author formulated the motivating mathematical idea and initial hypotheses, supplied the starting basis for the investigation, and pursued the research with AI assistance. The reported results were developed through this internally AI-assisted process. The author also supplied the publication identity and new runtime reports. This contribution statement records the author's account of his role in originating and developing the project; it makes no priority claim for the established mathematical methods used here.

Internal mathematical investigation and inspection were AI-assisted. Completed personal verification of every proof step is not asserted, and no independent human peer review of this draft is documented. AI tools are not authors.

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